Calibration-derived decoder discriminability is associated with online P300-speller accuracy, but the fitted mapping does not transport across cohorts

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Abstract

Objective

A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a fitted mapping from that score to expected accuracy transports to withheld cohorts.

Approach

Retrospective secondary analysis of BigP3BCI 1.0.0. Of 20 source studies, 18 yielded eligible online outcomes: 271 participants, 739 session-condition records, 19,611 character selections. The four documented amyotrophic lateral sclerosis (ALS) cohorts were the originally planned subgroup, before widening to every source study with an eligible online outcome. The predictor was calibration-derived decoder discriminability: cross-validated discriminability of a classifier fitted only to calibration epochs. One source study at a time was withheld from development. Between-cohort variation was summarised by the random-effects standard deviation tau, with participant-clustered standard errors.

Main Results

The association was positive in all 18 cohorts but varied widely in magnitude and precision (Pearson r 0.190 to 0.928; participant-level r = 0.714, $p < 0.001$). Pooled estimation error was 0.098 (95% CI 0.091 to 0.107) against a development-mean benchmark of 0.146. Calibration

did not transport: the intercept had tau 0.87, with a 95% interval for an unrepresented cohort of -1.97 to 1.85 on the log-odds scale, spanning mappings that badly understate and badly overstate accuracy. Slope heterogeneity led to the same conclusion, at tau 0.43 and an unrepresented-cohort interval of 0.111 to 2.005. Skill was negative in one of 18 cohorts. The four ALS cohorts alone gave an uncorrected between-cohort slope spread of 0.223, against 0.560 overall.

Significance

The score carries a reproducible signal about accuracy in the corresponding session, but the mapping between them is cohort-specific: usable for ranking sessions within a setting, not for reporting expected accuracy elsewhere without local recalibration. Few-cohort evaluation, as the ALS subgroup illustrates, understates how much performance varies elsewhere.

Introduction

The P300 speller lets a person select characters from a visual matrix using event-related potentials rather than movement, and it remains one of the few communication routes available when motor control is lost.[1,2] It has been evaluated repeatedly in amyotrophic lateral sclerosis (ALS),[3,4] including long-term independent use at home,[5,6] and it sits alongside implanted systems as part of the communication options for people with severe paralysis.[7,8]

Performance is not uniform. Online spelling accuracy varies widely between users and between sessions in the same user, and a proportion of users do not reach usable accuracy at all.[9] This variability motivated a line of work asking whether performance can be anticipated from data recorded before a session begins. Neurophysiological features measured at rest or during a short calibration block relate to later brain-computer interface performance,[10] and in ALS specifically, event-related potential and spectral measures recorded during calibration relate to P300-speller accuracy.[11,12] Attentional and motivational state have also been linked to performance in this population.[13,14] Mainsah and colleagues went furthest, deriving P300-speller accuracy analytically as a function of a calibration-derived detectability index, with the explicit purpose of estimating performance without extensive online testing.[15] A projected-accuracy measure built from flash-level data and a multi-feature predictor built from a separate rapid-serial-visual-presentation task have made related predictions from within-session or within-study data.[37,38]

Two features of that literature limit what it can support. First, the relationships were established within the cohort in which they were measured. A relationship that holds inside one study does not establish that a mapping fitted in one set of cohorts will estimate accuracy in a cohort it has never seen, which is the question that matters if a calibration score is to be used anywhere other than where it was developed. Second, these studies report discrimination, usually as a correlation or an area under the receiver operating characteristic curve. Discrimination describes whether users can be ranked. It does not describe whether the estimated accuracy is numerically close to the accuracy that follows, which is what a session-level check would require. To our knowledge, no published study has evaluated whether a calibration-derived score's fitted mapping to online accuracy transports to a source study withheld from its development, as distinct from evaluating the score's within-cohort association or its within-study discrimination.

Evaluating transportability also imposes a requirement that has not previously been met in this setting. When a model is evaluated by withholding an entire cohort, the cohort is the unit of replication, and the spread of performance across withheld cohorts is what determines the interval a reader should expect in their own setting.[16,17] Estimating that spread from a small number of cohorts yields an interval too imprecise to be informative. Public archives that aggregate many independent P300-speller studies under a shared recording montage now make a larger number of withheld cohorts available.[18] A recent cross-dataset evaluation aligned P300 event-related-potential signals across independent archives to improve classification transfer, a different question from whether a fitted calibration-to-accuracy mapping itself transports numerically.[39]

We evaluated whether a calibration-derived score estimates online session accuracy in the corresponding P300-speller session, in cohorts withheld from model development, across every source study in a public archive that yields eligible online outcomes. The score used here is the cross-validated discriminability of a classifier fitted to a session's calibration epochs, referred to throughout as calibration-derived decoder discriminability to keep it distinct from a physiological marker of user aptitude. The documented ALS cohorts were the originally planned subgroup, before the design widened to every source study with an eligible online outcome. Because the archive also contains cohorts without a documented ALS population, we further asked whether a mapping developed without any ALS data transports to the ALS cohorts, and whether the calibration-to-accuracy relationship itself differs between cohort types.

Methods

Study Design and Reporting

This was a retrospective secondary analysis of a public archive of legacy online P300-speller recordings. The design was an internal-external cross-validation across source studies, in which an entire source study was withheld from model development and the fitted mapping was evaluated in that withheld cohort.[16,17] It is not an external validation in the strict sense, because every cohort comes from one harmonised archive rather than from a separately assembled dataset; what is withheld is the source study, and the source study is what the transportability question is asked about. Reporting follows the TRIPOD statement for prediction-model studies.[19]

No analysis plan was registered. The four documented ALS cohorts were fixed as the originally planned subgroup in the project documentation before the design was widened beyond them. The sensitivity and comparator analyses were listed in the Methods before either was run on the widened cohort, but after the primary widened analysis had been fitted, so neither is prespecified in the sense that a registered plan would establish; the protocol-descriptor moderator analysis was added later still, and is exploratory for that reason as well as for the reasons given below. The commit history of the analysis repository records these dates.

Calibration and online blocks come from the same recording session throughout. Every timestamp in the archive is de-identified, so recording order cannot be established from file headers, and the separation enforced here is between protocol phases and between the data used to fit the predictor and the data used to measure the outcome, not a demonstrated temporal precedence. An analysis in which the predictor comes from a session preceding the outcome session is reported separately as a secondary result. Session order was taken from the lexical order of session identifiers, which the archive assigns sequentially within a participant. Recording timestamps are de-identified and cannot confirm this, so the preceding-session analysis rests on the assumption that identifier order matches recording order.

Data Source and Cohort

The source was BigP3BCI version 1.0.0.[18] The downloaded archive was fixed by SHA256 digest (`eea294aa34e9ed11e5a25d07e30aeefdf8b2d467a8309e2c38405a289afcd72f`) before ingestion, and every selected European Data Format file was checked against the distributor checksum manifest.

All 20 documented source studies were screened. Every study supplied the shared 16-channel montage sampled at 256 Hz together with the stimulus and phase event channels, so montage compatibility excluded none. Two studies contributed no eligible online outcome and could not enter the analysis: in one, every feedback phase was overridden by artificial feedback, and in the other the intended character could not be recovered in any feedback phase. Both are retained in the study inventory with their reason, because their exclusion follows from the archive rather than from any analysis choice.

The archive documentation identifies an ALS study population for four source studies (Study Design and Reporting, above). The remaining cohorts carry no documented ALS population and are described here as other cohorts rather than as healthy or control cohorts, because the documentation does not support a positive characterisation.

Participant identifiers are study-scoped; because the source studies draw from a limited number of contributing laboratories, participant overlap across studies cannot be ruled out from the archive's documentation (supplement, S1).

Calibration Predictor

Only Train-phase files informed the predictor. A calibration event was a rising `StimulusBegin` transition during phase 2, labelled target or non-target by `StimulusType`. The 16 shared channels were filtered from 0.5 to 30 Hz with a fourth-order zero-phase Butterworth filter, epoched from 200 ms before to 800 ms after each event, baseline corrected to the prestimulus interval, and rejected when absolute amplitude exceeded 150 microvolts. A session entered the analysis only if at least 10 target and 40 non-target epochs survived rejection.

Each epoch was decimated by taking every fourth sample of the 256 Hz signal, giving 64 samples per channel and 1,024 features at an effective rate of 64 Hz. The factor was chosen so that the resulting Nyquist frequency of 32 Hz stays above the 30 Hz upper edge of the bandpass, which keeps the retained band from folding onto lower frequencies. Extraction rejects any file sampled below 240 Hz, the rate at which this factor would begin to fold the passband.

The primary score, referred to throughout as calibration-derived decoder discriminability, was the out-of-fold area under the receiver operating characteristic curve of an L2-regularised logistic classifier (C = 1.0, lbfgs solver) trained on standardised features, with stratified grouped cross-validation grouping by European Data Format file so that no epoch was scored by a model fitted on the same recording file. The number of folds was the smaller of five and the number of files in the session, so fold count varies between sessions and studies. The score is a property of a session and is constant across the conditions recorded within it. The quantity is a measure of how well that session's calibration data support a decoder, not an independent physiological marker of user aptitude.

The regularised logistic classifier and the linear discriminant comparator were chosen because they are the families evaluated for this paradigm in the P300-speller literature.[20,21] Spatially filtered and Riemannian pipelines,[22,23,24] and the wider set of classifiers reviewed for event-related potential decoding,[25] were not used, because the study evaluates a mapping from a calibration summary to online accuracy rather than proposing a decoder.

Comparator scores computed from the same calibration epochs were a regularised linear discriminant analysis area under the curve, grouped cross-validated classification accuracy, mean target-minus-non-target amplitude at Pz between 250 and 500 ms, the same contrast averaged over six posterior channels, and the maximum posterior signed r-squared. Each was run through the identical withheld-cohort procedure and its results are reported in the supplement. An exploratory specification adding ALSFRS-R to the primary score is restricted to the records carrying an observed value, which is 3 of the 18 cohorts, so it is reported alongside the primary score run on those same records rather than against the full archive.

Outcome

At each Test-phase transition to phase 3, the intended character was the final nonzero `CurrentTarget` in the directly preceding phase-2 interval and the selected character was the modal nonzero `SelectedTarget` during phase 3. A feedback phase was eligible when feedback was displayed, both values could be recovered, and artificial feedback did not override the selection. Correctness required an exact character match. The outcome was the proportion of correct eligible selections within a participant-session-condition record.

Model Specification

The development model was a logistic regression of character-level correctness on the standardised calibration score, fitted on character-expanded records from the development studies. Standardisation used development-study means and standard deviations only. For a session with calibration score s, the estimated accuracy is the inverse logit of a + b (s - m) / d, where m and d are the development mean and standard deviation of the score and a and b are the fitted coefficients; fitted values for every fold are reported in the supplement (Table

S9) so that any estimate can be recomputed. Because records are expanded to one row per character, a session contributes to the fit in proportion to the characters it supplied, and the quantity estimated is the accuracy of a randomly chosen character selection. Refitting the same withheld-cohort evaluation with each session weighted equally, and again with each participant weighted equally, moved the pooled estimation error by less than 0.003, so the reported error does not rest on that choice (supplement, S3).

Validation Design

The primary evaluation withheld one source study at a time. Three further analyses used the wider cohort set. The ALS subgroup analysis repeated the same procedure within the four ALS cohorts alone. The transfer analysis developed the mapping on the other cohorts only, with every ALS cohort withheld simultaneously, and evaluated it in each ALS cohort. The moderation analysis tested whether the cohort-specific calibration slope differed between ALS and other cohorts. That slope is the coefficient of the model's predicted logit in a binomial regression of observed correctness fitted within each withheld cohort, with standard errors clustered on participant. Cohort type does not vary between the records of a cohort, so the cohort is the unit of that comparison and a record-level interaction would treat a study-level attribute as though it had been measured once per record. Because four cohorts in one group cannot support a single decisive test, the comparison is reported three ways: Welch's two-sample t-test on the 18 cohort-specific slopes, which uses no weighting and is the primary form; an exact permutation test that enumerates all 3,060 ways of assigning the ALS label to four of the 18 cohorts and studentises the contrast with the same Welch statistic, so that it does not depend on the ratio of the two groups' variances; and a random-effects meta-regression in which each cohort is weighted by the inverse of the sum of its own sampling variance and the residual between-cohort variance, the latter estimated by the method of moments with cohort type in the model, with the Knapp and Hartung adjustment for the small number of cohorts. The comparison was refitted with each cohort dropped in turn.

Statistical Analysis

The primary metric was the mean absolute difference between estimated and observed session-condition accuracy in the withheld cohort. Because an absolute error is not interpretable without a reference, three references are reported: a development-mean benchmark that estimates every withheld record at the development-set mean accuracy, a held-out-cohort-mean benchmark that estimates every withheld record at that cohort's own mean accuracy, and the skill of the model against the development-mean benchmark, defined as one minus the ratio of their errors. Skill is reported against the development-mean benchmark throughout unless the cohort's own mean is named. The second benchmark is the harder of the two pooled, but not in every cohort: mean absolute error is minimised by the median rather than the mean, so a cohort's own mean is not guaranteed to beat any other constant, and it is the easier target in 7 of the 18 cohorts. Secondary metrics were root mean squared error, character-weighted Brier score, Brier skill score against development prevalence, calibration intercept and slope, and character-weighted area under the curve. Calibration intercept and slope are reported per withheld cohort as well as pooled, because a pooled value can conceal opposing departures in individual cohorts.[26,27]

Two uncertainty statements are reported and are not interchangeable. Confidence intervals from 2,000 deterministic bootstrap replicates, in which development participant clusters were resampled within source study, the model was refitted, and withheld participant clusters were resampled, describe uncertainty conditional on the observed set of source studies. Separately, the source study was treated as the unit of replication: the withheld-cohort estimates were summarised by their mean and between-study standard deviation, with a t-distributed confidence interval for the mean and a prediction interval for a cohort not represented in the archive.[16,17] Cohort-level mean absolute error was pooled after natural-log transformation, because the quantity cannot be negative; the reported mean is the back-transformed geometric mean, and its between-cohort standard deviation is reported on the log scale, not back-transformed. The prediction interval is the quantity a reader should use when asking what to expect in their own cohort.

For the calibration intercept and slope, which are the quantities that carry the transportability question, that summary was replaced by a random-effects meta-analysis of the 18 cohort estimates, because the cohorts differ in size and the uncorrected standard deviation of the estimates counts their sampling error as though it were between-cohort variation. The between-cohort standard deviation tau was estimated by the method of moments[28] with a Q-profile confidence interval,[29] the Paule and Mandel estimate[30] is reported alongside it so that the estimator choice is visible, and the Q test, I^2[31] and a 95% prediction interval for an unrepresented cohort accompany each. All 18 cohorts were fitted under the cluster, model and quasi-binomial specifications and none was dropped. Under the participant-cluster bootstrap check reported in the Discussion, Study S1's cluster resamples fell onto the separation boundary often enough that fewer than half of its 2,000 replicates left an identified estimate, and that cohort is reported as not identified under the bootstrap method alone. Because the bootstrap's smaller tau could reflect either the estimator or the missing cohort, the cluster-robust pooling was repeated on the identical 17 cohorts the bootstrap identified, excluding Study S1: tau = 0.44 for the slope and 0.87 for the intercept, no closer to the bootstrap's own 0.37 and 0.77 than the all-18 cluster-robust values of 0.43 and 0.87 were, so most of the difference between the two methods reflects the variance estimator rather than Study S1's exclusion (supplement, S4).

The primary specification clusters each cohort's standard errors on participant, because the 739 records come from 271 participants and the conditions recorded within one session share a predictor value by construction, so a binomial likelihood that treats each record as an independent draw understates the within-cohort variance and thereby inflates the estimated between-cohort variance. A quasi-binomial specification, which scales the same standard errors by the square root of the Pearson dispersion, and the uncorrected model-based specification are reported as sensitivities. The cluster-robust and quasi-binomial results agree at the pooled level, though not cohort by cohort, where their standard errors differ by up to 40% and clustering shrinks the standard error relative to the model-based fit in 3 of 18 cohorts.

Each cohort contains between 5 and 24 participants, and a cluster-robust variance is downward-biased below roughly 30 clusters even with the finite-sample correction applied here. That bias inflates tau and I^2, so the I^2 reported below is an upper estimate: a 20% understatement of the within-cohort variances would put the slope I^2 below the conventional 75% threshold. I^2 is therefore reported with its confidence interval as a description, and no conclusion is drawn from its position relative to that threshold. The conclusion is instead checked by refitting at each confidence limit of tau and by repooling with every within-cohort variance multiplied by a common factor.

Protocol descriptors were tested as moderators of the cohort calibration slope within the same random-effects framework, each cohort weighted by the inverse of its own variance plus the residual between-cohort variance, the moderator standardised so that its coefficient reads per between-cohort standard deviation, and inference by the Knapp and Hartung adjustment on k minus two degrees of freedom. Ten were available. The median inter-selection interval was derived from the reconstructed selection timestamps as the difference between consecutive selections within a single recording file, summarised as a median per file and then as the median of those file medians; differences were never taken across a file boundary, because files are separate recordings whose de-identified timestamps share no origin, and the first selection of each file contributes none. It was computed over all reconstructed selections rather than the eligible ones alone, because the pacing of a recording does not depend on which of its selections survived eligibility screening, and a cohort was classed as fixed-interval when the relative spread of that interval within a file was at most 1e-6, a cut that separates a spread of at most 8e-16 from one of at least 0.050 and so is not a threshold the data are sensitive to. The largest target index and the number of distinct targets were taken over the same trials as empirical proxies for the size of the speller matrix, bounded below by the alphabet actually spelled; grid size and stimulus paradigm, in contrast, were transcribed directly from the archive's own data descriptor, which documents both per source study. A study was coded as using a checkerboard paradigm if any of its documented paradigms was a checkerboard variant (CB, CBcol, or sCB), because those variants share the flash-adjacency-avoidance design the checkerboard paradigm was built for; only 2 of the 18 contributing studies (Study D, Study J) lacked one. The archive's EDF+ header dictionary also documents an equipment code, but as a single value (gUSBAmp) applying to the entire archive rather than a per-study field, so it does not vary across cohorts and could not be entered as a moderator. The number of stimulus conditions and the median selections per record come from the eligible trials and the analysis records; and mean accuracy, its between-record standard deviation and the fraction of records at ceiling come from the analysis records. P values were corrected across the ten descriptors by the Holm step-down procedure, which is valid under any dependence between the tests, within each of the two analyses separately. The share of the between-cohort variance a moderator accounts for is one minus the ratio of the residual tau squared with that moderator to the same estimator without it, and is reported as zero where that ratio exceeds one.

Because records are session-conditions nested within sessions within participants, and because the predictor is constant within a session, the intraclass correlation of session accuracy within participant is reported alongside an effective number of independent sessions. Associations are reported as Pearson and Spearman coefficients at the session, study-centred and participant levels. Precision for the calibration slope is reported as the departure of the pooled slope from unity that a Wald z test of the pooled slope against unity could have detected with 80% power at a two-sided 5% threshold, computed as 2.80 times the standard error of the pooled slope under the primary cluster-robust random-effects specification. That number concerns the average slope across cohorts and not its between-cohort spread, which is the quantity a single fitted mapping has to survive and which is reported as tau with its confidence interval. Four sensitivity analyses, all listed before any was run on the widened cohort, covered restriction to cohorts with meaningful outcome variance, exclusion of records with high artifact rejection, exclusion of small-denominator records, and leave-two-studies-out development. In the last of these every pair of cohorts was withheld together, so that development ran on 16 cohorts rather than the 17 of the primary analysis. Each cohort appears in 17 of the 153 pairs, and its 17 estimates were averaged into a single value before the same across-cohort pooling was applied, because a pair shares a cohort with 32 other pairs and the pairs are therefore not independent units.

The significance threshold was $P < .05$, two-sided. Analyses used Python 3.11 with NumPy 2.3, SciPy 1.16, scikit-learn 1.7, statsmodels 0.14, and pandas 2.3.

Ethics

This was a secondary analysis of BigP3BCI version 1.0.0 (doi:10.13026/0byy-ry86), a publicly available, fully de-identified archive. It involved no new data collection, no participant contact, and no intervention, and institutional review board approval was therefore not required. The original source studies' ethics approvals and consent statements are reported in the archive documentation.

What the Calibration Score Is, and What It Is Not

In these copy-spelling protocols the Train phase supplies the data from which the online classifier is derived before the Test phase begins, and the source-study publications that report this practice describe a linear classifier fitted on each session's calibration data.[32,33] The archive itself names no online classifier for any source study, and the BCI2000 parameter files that would carry its specification were withheld from the release, so the deployed decoder cannot be reconstructed here. The predictor evaluated in this study is in any case not that decoder: it is the out-of-fold discriminability of a separate classifier fitted here on the same calibration recordings. The score is therefore reported as the cross-validated learnability of that session's calibration data, not as a property of the deployed decoder and not as a physiological marker of user aptitude.

Results

Cohort

All 20 documented source studies supplied the shared 16-channel montage at 256 Hz and were screened. Eighteen contributed at least one eligible online outcome. One study contributed none because artificial feedback overrode the selection in all 5,680 reconstructed feedback phases, and one contributed none because the intended character could not be recovered in any of 2,263 phases (Table 1).

The analytic set contained 271 study-scoped participants, 410 participant-sessions, 739 participant-session-condition records, and 19,611 online character selections, of the 19,688 eligible selections in Table 1. The 77 selections that separate the two totals are the two Study B sessions archived without a calibration block, described in the Table 1 legend and reconciled cohort by cohort in Table S1. The four documented ALS cohorts contributed 47 participants, 113 sessions, 194 records, and 3,318 selections. Observed session-condition accuracy averaged 0.851 and 228 of 739 records (30.9%) were at 100%, with three cohorts near ceiling (mean accuracy 0.963 to 0.997).

Table 1. Source studies screened, and their contribution to the analytic set. All 20 documented source studies supplied the shared 16-channel montage at 256 Hz. Two contributed no eligible online outcome: Study C, in which artificial feedback overrode the selection in all 5,680 reconstructed feedback phases, and Study P, in which the intended character could not be recovered in any of 2,263 phases. Cohorts are marked according to whether the archive documentation identifies an ALS study population. The two totals in the final rows differ by 77 selections, all of them in Study B: two sessions of one Study B participant were archived with Test-phase recordings only and carry no calibration block at all, so no calibration score exists for them and their 38 and 39 selections cannot enter a model that takes that score as its predictor. Every other eligible selection in the archive is analysed, and the per-cohort selection counts in Table 2 are the analysed counts.

Source study	ALS documented	Phases reconstructed	Eligible selections	Contribution
Study A	No	1,404	1,404	Contributed
Study B	Yes	858	858	Contributed
Study C	No	5,680	0	Artificial feedback overrode every selection
Study D	No	1,230	1,230	Contributed
Study E	No	240	240	Contributed
Study F	Yes	1,079	1,067	Contributed
Study G	No	1,198	1,198	Contributed
Study H	No	1,926	1,926	Contributed
Study I	No	948	948	Contributed
Study J	No	1,812	1,812	Contributed
Study K	No	480	480	Contributed
Study L	Yes	990	990	Contributed
Study M	No	1,260	1,260	Contributed
Study N	Yes	480	480	Contributed
Study O	No	1,202	1,187	Contributed
Study P	No	2,263	0	Intended character not recoverable
Study Q	No	3,888	1,944	Contributed
Study R	No	1,440	1,440	Contributed

Source study	ALS documented	Phases reconstructed	Eligible selections	Contribution
Study S1	No	360	360	Contributed
Study S2	No	864	864	Contributed
Total eligible feedback phases	4 of 20 cohorts	**29,602**	**19,688**	18 cohorts contributed
Analysed selections	4 of 18 cohorts	-	**19,611**	after requiring a usable calibration feature set

Because the calibration score is a property of a session, the 739 records carry 410 distinct predictor values. Session accuracy clustered within participant (intraclass correlation 0.412), giving approximately 338 effective independent sessions.

Association Between Calibration-Derived Decoder Discriminability and Online Accuracy

The point estimate was positive in all 18 contributing cohorts, but its magnitude varied widely and several cohort-specific estimates were imprecise, with confidence intervals including values close to zero (Table S3); pooled across sessions the association was precisely estimated, at r = 0.677 (95% CI 0.621 to 0.726) once both variables are centred within cohort. Within-cohort Pearson r ran from 0.190 to 0.928 (median 0.635), and the 95% confidence interval included zero in 6 of the 18 cohorts; the widest, in a cohort of eight sessions, ran from -0.42 to 0.86. Pooling sessions gave $r = 0.716$ (95% CI 0.665 to 0.760, $n = 410$, $p < 0.001$). Centring both variables within cohort, which removes cohort means though not differences in variance, protocol or measurement precision, gave $r = 0.677$ (0.621 to 0.726, $p < 0.001$), so the association is not produced by cohorts differing in mean difficulty and mean signal quality. Collapsing each participant to a single observation gave $r = 0.714$ (0.650 to 0.768, $n = 271$, $p < 0.001$; Spearman $\rho = 0.756$).

Estimation Error and Its Reference Points

Pooled across withheld cohorts, estimated session accuracy fell about a third closer to observed accuracy than the development-mean benchmark, and also beat the benchmark of each cohort's own mean. Across withheld cohorts the mean absolute error was 0.098 (95% CI 0.091 to 0.107), against 0.146 for the development-mean benchmark and 0.123 for the held-out-cohort-mean benchmark, giving a skill of 0.327 against the development-mean benchmark. This number pools every withheld record and so weights the larger cohorts more heavily. The mean of the 18 cohort-level errors, reported next, is a different quantity and the two are not interchangeable, and the per-cohort picture is less uniform than the pooled one. The character-weighted Brier score was 0.123 (95% CI 0.113 to 0.132), Brier skill 0.110 (95% CI 0.063 to 0.149), and character-weighted area under the curve 0.748 (95% CI 0.719 to 0.770). These intervals are conditional on the 18 observed cohorts.

Transportability

Treating the source study as the unit of replication, discrimination transported on average but not dependably, and calibration did not transport (Figures 1 and 2). Across the 18 withheld cohorts the mean absolute error, pooled on the log scale because the quantity cannot be negative, averaged 0.090 (the geometric mean; the arithmetic mean of the 18 cohort-level errors was 0.101) with a between-cohort standard deviation of 0.477 on the log scale, giving a 95% interval for the mean of 0.071 to 0.115 and a 95% interval for a cohort not represented in the archive of 0.032 to 0.254 (Figure 3). Area under the curve averaged 0.713 across cohorts, with a new-cohort interval of 0.491 to 0.935, the lower bound of which is chance. Brier skill averaged 0.174 across cohorts with a new-cohort interval of -0.296 to 0.644. Every study-level summary in this paragraph is tabulated in Table S2.

Figure 1. Calibration slope and intercept in each withheld cohort, against the values a transportable mapping would take. Each row is one cohort withheld from model development, with its estimate and 95% confidence interval, marked by whether the archive documents an ALS study population. Standard errors are clustered on participant. Each panel is ordered by its own estimate, so a cohort does not occupy the same row in panel a as in panel b. The dashed line is the value a mapping that transported exactly would take, one for the slope in panel a and zero for the intercept in panel b; the band is the 95% interval for a cohort not represented in the archive and the solid line within it is the pooled estimate. That band is computed at the point estimate of tau, which is itself imprecise, so it widens and narrows with the values of tau reported in the Results. Intervals running past the axis are drawn with an arrow. The between-cohort standard deviation tau and I^2 are given beneath each panel; I^2 is descriptive only, for the reason given in the Results.

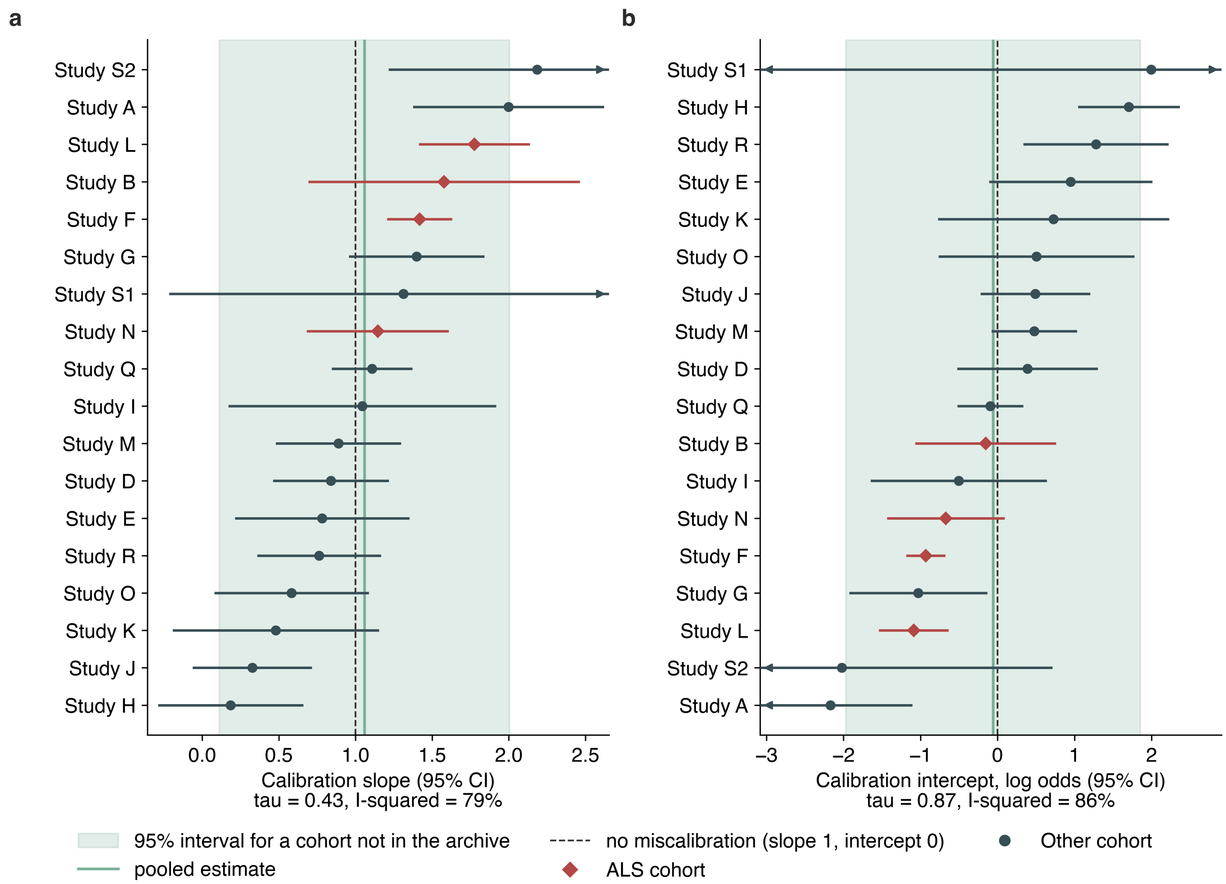


Figure 2. Observed against estimated session accuracy, in the four ALS cohorts and the two non-ALS cohorts at either end of the calibration-slope range (Figure 1). Each point is one of up to ten equal-count bins of the estimate within that cohort, placed at the selection-weighted mean estimate and the observed accuracy of the records in it, with point area increasing with the selections it rests on. The dashed line is equality. Points above it are bins whose accuracy the mapping understated and points below are bins whose accuracy it overstated. The number in each panel is observed minus estimated accuracy across that whole cohort. The full 18-cohort version of this figure is Figure S2.

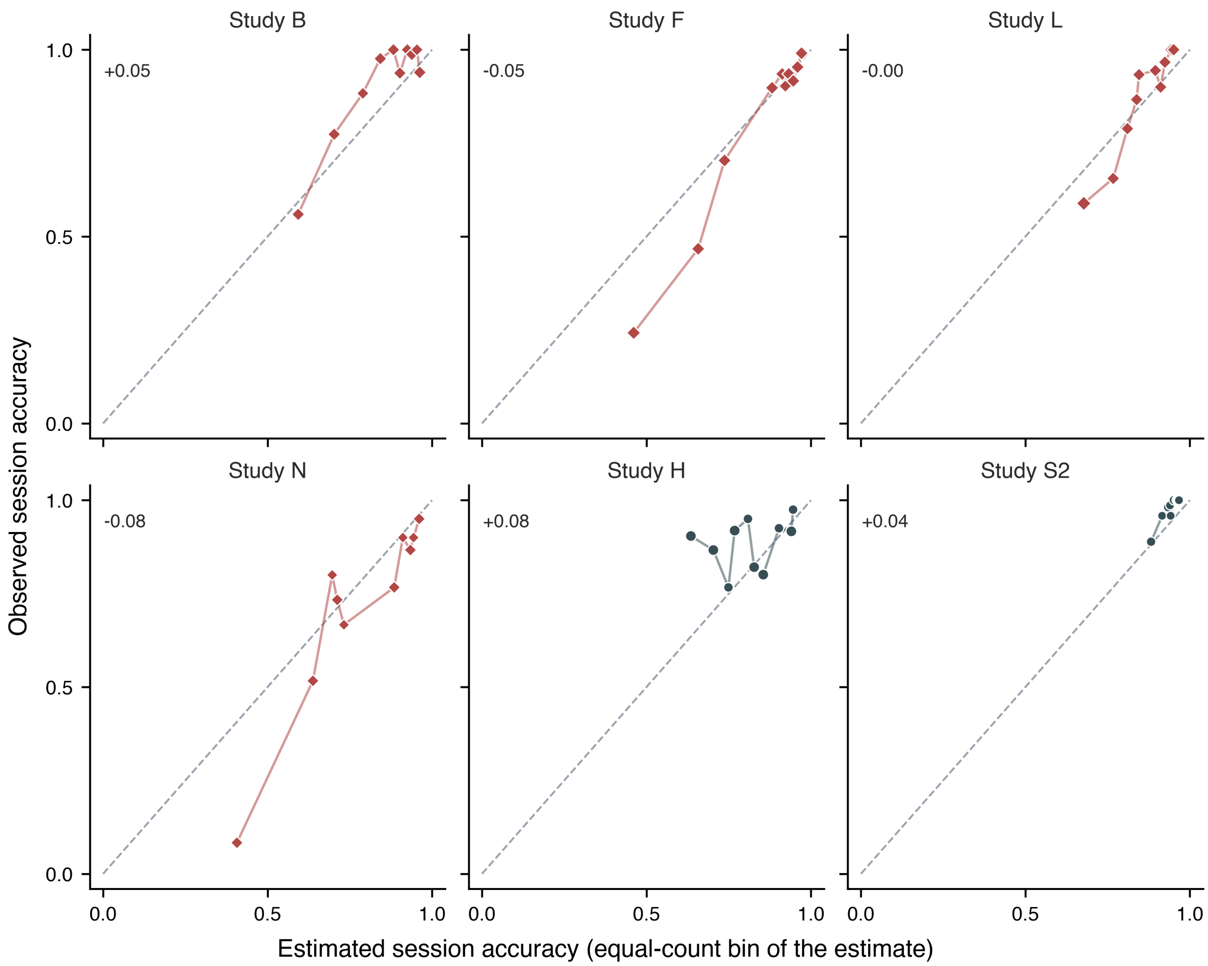
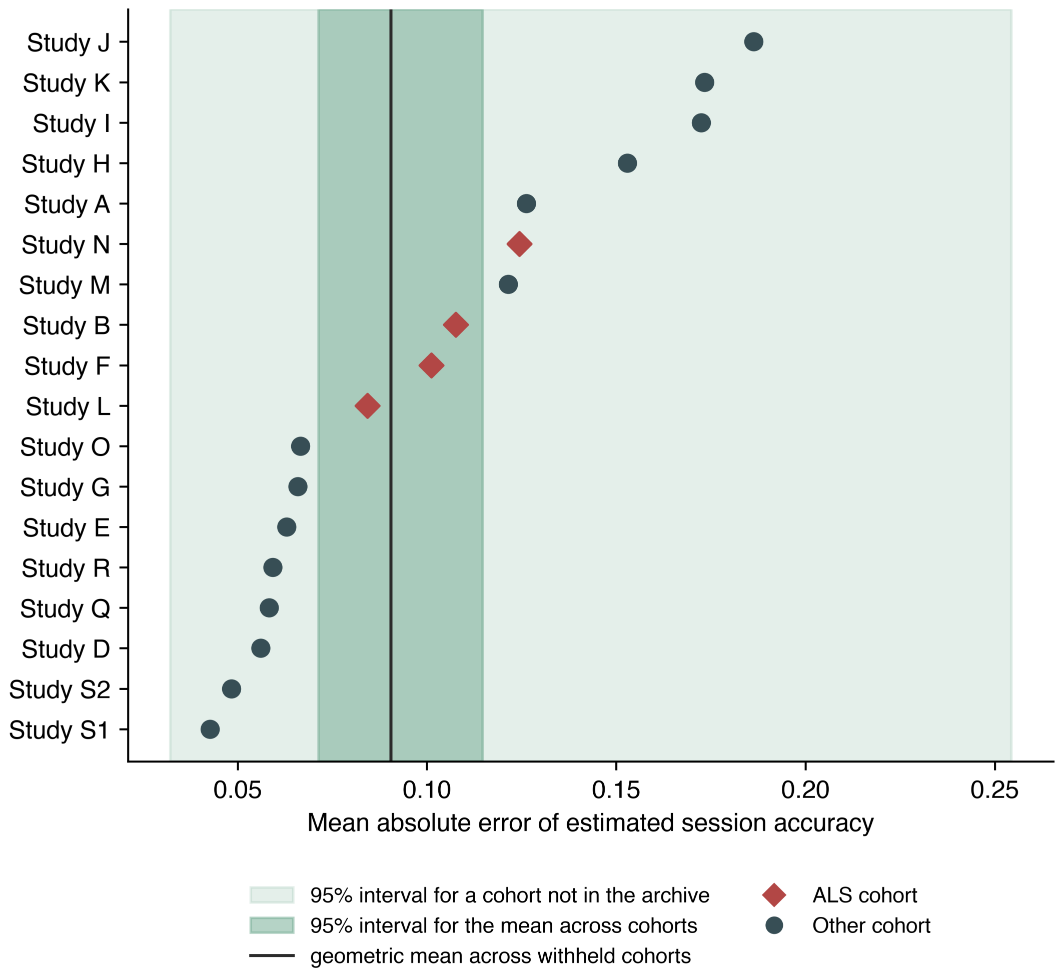


Figure 3. Estimation error in each withheld cohort, with the two uncertainty statements. Each point is one cohort withheld from model development. The darker band is the 95% interval for the geometric mean across the observed cohorts; the lighter band is the 95% interval for a cohort not represented in the archive. The two answer different questions and only the second describes what a reader should expect in their own setting.



The calibration intercept varied so widely between cohorts that a single fitted mapping was displaced in many of them. The intercept departed from zero by at least half a log-odds unit in 13 of the 18 cohorts and by at least a full unit in 7, and its 95% interval excluded zero in 6. All 18 cohorts yielded an identified intercept and slope and none was dropped from the pooling. Under the primary cluster-robust specification the between-cohort standard deviation of the calibration intercept was tau = 0.87 (95% CI 0.60 to 1.51) against a pooled intercept of -0.06, and Q = 122.6 on 17 degrees of freedom ($p < 0.001$). I^2 was 86.1 (95% CI 79.5 to 90.6), reported as a description only: it is an upper estimate at these cluster counts, and a 20% understatement of the within-cohort variances would put its lower confidence limit below the conventional 75% threshold. The observed intercepts ran from -2.169 to 1.995, and the 95% interval for a cohort not represented in the archive runs from -1.97 to 1.85 on the log-odds scale (Figure 1), which is the difference between a mapping that badly understates accuracy and one that badly overstates it. That conclusion does not depend on where in its interval tau lies: setting tau to its own lower 95% confidence limit still leaves a new-cohort interval of -1.39 to 1.26, and the interval is still -1.37 to 1.06 when every within-cohort variance is inflated fivefold.

The calibration slope varied in the same direction, and its between-cohort variance is itself imprecisely estimated. tau = 0.43 (95% CI 0.30 to 0.77) against a pooled slope of 1.06 (95% CI 0.82 to 1.30), with Q = 81.3 on 17 degrees of freedom ($p < 0.001$) and I^2 79.1 (95% CI 67.6 to 86.5), an interval that spans that same threshold, so no claim here rests on it either. The observed slopes ran from 0.185 to 2.185 and the 95% interval for an unrepresented cohort runs from 0.111 to 2.005 (Figure 1); at the lower confidence limit of tau it is 0.394 to 1.704, across which no single fitted mapping is usable. That interval excludes zero at the point estimate of tau but includes it at tau's upper confidence limit, where it runs from -0.603 to 2.759, so the exclusion is a property of this interval at one value of tau rather than a finding. Cohort by cohort, 4 of the 18 slope confidence intervals include zero (Figure 1), so the association is reliable on average and not reliable in every cohort. Fitting one calibration curve to all withheld predictions pooled gives a slope of 0.967 (95% CI 0.791 to 1.127) and an intercept of 0.054 (95% CI -0.264 to 0.395); those pooled values average opposing departures and describe no individual cohort, and the cohorts whose accuracy the mapping understated and those whose accuracy it overstated are visible separately in Figure 2.

Table 2. Per-cohort composition and observed performance. For each of the 18 contributing cohorts: participants, records, analysed selections, mean observed session-condition accuracy, and, when that cohort was withheld from model development, the mean absolute error and character-weighted area under the curve. ALS cohorts are listed first. The calibration intercept and slope for each cohort, with their 95% confidence intervals, are shown in Figure 1 rather than tabulated here; the full per-cohort table, which also carries the archive's documented grid size and checkerboard-paradigm indicator for each cohort alongside the calibration intercept and slope, is Table S11 in the supplement.

Cohort	Participants	Records	Selections	Observed accuracy	MAE	AUC
Study B	18	56	781	0.899	0.108	0.825
Study F	10	89	1,067	0.778	0.101	0.863
Study L	11	33	990	0.839	0.084	0.811
Study N	8	16	480	0.694	0.124	0.814
Study A	13	39	1,404	0.786	0.126	0.775
Study D	17	34	1,230	0.896	0.056	0.641
Study E	8	8	240	0.921	0.063	0.668
Study G	20	40	1,198	0.886	0.066	0.732
Study H	16	64	1,926	0.877	0.153	0.520
Study I	13	26	948	0.658	0.172	0.674
Study J	20	40	1,812	0.708	0.186	0.600
Study K	5	16	480	0.750	0.173	0.639
Study M	21	42	1,260	0.823	0.121	0.685
Study O	17	34	1,187	0.884	0.067	0.581

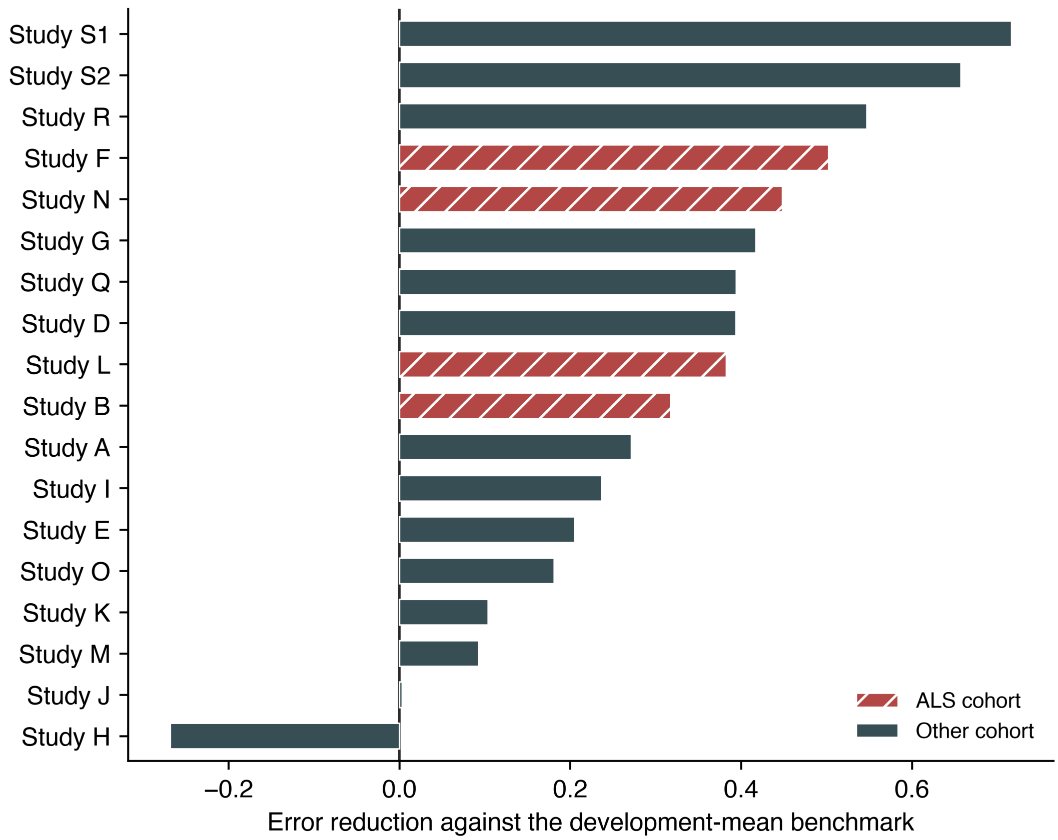
Cohort	Participants	Records	Selections	Observed accuracy	MAE	AUC
Study Q	20	54	1,944	0.820	0.058	0.682
Study R	20	80	1,440	0.963	0.059	0.650
Study S1	10	20	360	0.997	0.043	0.851
Study S2	24	48	864	0.976	0.048	0.826

A small average departure of the slope from unity was not within reach of this design. The pooled slope carries a standard error of 0.122 under the primary specification, so a Wald z test of that slope against unity would have detected a departure of at least 0.34 with 80% power at a two-sided 5% threshold, and its confidence interval has a half-width of 0.24; departures smaller than that cannot be excluded. Neither number bears on transportability, because what a single fitted mapping has to survive is the spread of the slope across cohorts rather than its average, and for that quantity the design placed tau between 0.30 and 0.77 without resolving where in that range it lies.

The uncorrected standard deviations of the 18 cohort estimates, tabulated in the supplement (Table S2), are 0.560 for the slope and 1.175 for the intercept. Both exceed tau because they count each cohort's sampling error as though it were between-cohort variation, and tau is used in preference throughout.

One cohort was estimated less accurately than the development-mean benchmark, and six were estimated less accurately than their own cohort mean. Skill against the development-mean benchmark was negative in one of 18 cohorts and positive in the remainder (Figure 4). Against the benchmark of each cohort's own mean, which is the harder target pooled and in 11 of the 18 cohorts, skill was negative in six of 18 (Table S5). Three distinct things produce those six. In Study H the model was less accurate than both benchmarks, and it is the same cohort that fails the development-mean benchmark. Four are cohorts whose observed accuracy sits high enough that their own mean already tracks nearly every record: in Study E, Study R, Study S2 and Study S1, accuracy averages 0.921, 0.963, 0.976 and 0.997 and the own-mean benchmark errs by 0.049, 0.047, 0.035 and 0.005, leaving little error for any predictor to reduce. Study S1 is the limiting case, where that near-zero denominator makes the ratio large and negative and not comparable with the rest. In the sixth, Study J, the shortfall is -0.008, which is parity rather than failure. All four ALS cohorts had positive skill against both benchmarks, at 0.317, 0.383, 0.448 and 0.502 against the development mean.

Figure 4. Error reduction in each withheld cohort against the development-mean benchmark. Values below zero indicate that the calibration score estimated accuracy less well than estimating every record at the development-set mean. All four ALS cohorts were positive; one cohort without a documented ALS population was negative.



Amyotrophic Lateral Sclerosis Subgroup

Restricting the analysis to the four ALS cohorts reproduced a more favourable and considerably more precise picture than the full archive supported. Within the originally planned subgroup the mean absolute error was 0.091 (95% CI 0.076 to 0.108), Brier skill 0.303 (95% CI 0.184 to 0.424), area under the curve 0.831 (95% CI 0.768 to 0.869), calibration intercept 0.025 (95% CI -0.240 to 0.341), and slope 0.985 (95% CI 0.786 to 1.188). The uncorrected between-cohort standard deviation of the calibration slope was 0.223 within this subgroup against 0.560 across all contributing cohorts. Four cohorts are too few to estimate tau usefully, so the uncorrected spread is used on both sides of that comparison.

Transfer From Cohorts Without a Documented ALS Population

A mapping developed without any ALS data estimated accuracy in the ALS cohorts with modest loss. With all four ALS cohorts withheld from development simultaneously, mean absolute error in each was 0.087, 0.106, 0.109, and 0.131 (mean 0.108), against 0.104 when other ALS cohorts were available for development. Signed bias ranged from -0.086 to 0.049.

Cohort Type as a Moderator

With the cohort as the unit of analysis, calibration slope was higher in the ALS cohorts, but not precisely enough to support a claim. The cohort-specific calibration slope averaged 1.479 in the four ALS cohorts (standard deviation 0.266) and 0.992 in the other 14 (standard deviation 0.580), a difference of 0.487 (standard error 0.204; 95% CI 0.040 to 0.933; Welch $t = 2.38$, $df = 11.7$, $p = 0.035$). An exact permutation test, which enumerates all 3,060 ways of labelling four of the 18 cohorts and assumes no distribution, gave $p = 0.058$. A random-effects meta-regression weighting each cohort by its own precision gave a difference of 0.559 (95% CI -0.009 to 1.127, $p = 0.053$), leaving a residual between-cohort standard deviation of 0.349 against 0.432 with cohort type removed, so cohort type accounts for about a third of the between-cohort variance in the slope and most of that variance remains unexplained.

The comparison also turns on individual cohorts. Dropping one ALS cohort moves the Welch p value to between 0.008 and 0.111, and dropping one of the other 14 moves it to between 0.013 and 0.060. The three tests agree in direction and in magnitude and fall on either side of the conventional threshold, and no one of them should be read as settling the question. A slope above one means observed accuracy varied more widely than the estimates did, so in the ALS cohorts the mapping compressed the range it should have spanned. These are the slopes of the primary leave-one-cohort-out model within each cohort, not the pooled slope of 0.985 reported for the ALS subgroup above, which comes from a model developed within the ALS cohorts alone. For the same reason their standard deviation of 0.266 is not the 0.223 reported there: 0.266 is the spread of the four ALS cohorts under the primary design and is what the per-cohort slopes in Table S11 sum to, and it is the number that belongs beside the 0.580 of the other 14, because both come from the same development set.

Cohort type is a study-level attribute that is entangled with paradigm, hardware and stopping rule, only four studies carry a documented ALS population, and the remaining cohorts are not documented as healthy controls. This comparison is exploratory and cannot separate population from protocol. The records underlying it are shown by cohort type in Figure S1, without a fitted line, for the reason given in the Methods.

Protocol Descriptors as Moderators

The time a cohort took over each selection was associated with a two-thirds reduction in estimated residual between-cohort variance in calibration slope. Ten descriptors were recoverable for every cohort and entered one at a time as moderators in a random-effects meta-regression of the cohort slope, weighting each cohort by its own precision and standardising the descriptor so that the coefficient reads per between-cohort standard deviation, with p values corrected across the ten by the Holm procedure. All 18 cohorts were fitted for every descriptor and none was dropped. The strongest was the median interval between consecutive selections within a recording, which ran from 7.3 to 50.0 seconds across cohorts and is the archive's only trace of the stopping rule, since a cohort that averaged more stimulus repetitions before committing to a character spent longer on each one: 0.406 per standard deviation (95% CI 0.181 to 0.632; $t = 3.82$, $df = 16$, $p = 0.002$, Holm $p = 0.015$), associated with a reduction in the between-cohort variance of the slope from tau squared = 0.187 to 0.063, a share of 66%, equivalently a residual between-cohort standard deviation of 0.25 against 0.43. The rank correlation agreed at rho = 0.72 ($p < 0.001$, Holm $p = 0.007$), and on the log scale, which is the more natural form for a duration, the share was 71% ($p < 0.001$). It is also the most stable result in this analysis: across the 18 leave-one-cohort-out refits its uncorrected p value never exceeded 0.010 and its share ran from 54% to 82%. Cohorts that spent longer per selection had steeper slopes. The reported effect of stimulus repetition is on accuracy rather than on the slope,[32,34,35] and the interval does not track cohort mean accuracy here (rho = 0.16, $p = 0.52$), so the direction is not read off that work. A mechanism is available, in that accumulated evidence grows with the square root of the number of repetitions, so a fixed difference in single-trial discriminability produces a larger difference in accumulated discriminability where repetitions are many, which would steepen the mapping from the score to accuracy. That argument is not tested here.

No other descriptor was associated with a share of the between-cohort variance distinguishable from zero, including the archive's own documented grid size and its documented use of a checkerboard-variant paradigm, alongside both empirical proxies for matrix size (supplement, S6). The result was also checked for robustness to how the stopping rule's descriptor was defined, to whether it is partly a consequence of the recording rather than a fixed protocol setting, and to how the multiple-comparison correction was applied across the ten descriptors tested; all three checks left the result materially unchanged (supplement, S6).

Predictor From a Preceding Session

Using a calibration recording from a preceding session, rather than from the session being estimated, attenuated but did not remove the association. Among 139 consecutive session pairs, the earlier session's calibration score correlated with the later session's accuracy at $r = 0.513$ (95% CI 0.379 to 0.626, $p < 0.001$), against $r = 0.728$ (0.639 to 0.798) for the score recorded in the session being estimated.

Sensitivity Analyses

Results were consistent under the four checks named in the Methods: excluding low-count records, restricting to cohorts with meaningful accuracy variation, excluding records with heavy calibration-epoch rejection, and leave-two-studies-out development, which withheld two cohorts at a time so that development ran on 16 rather than 17 cohorts and reproduced the primary result across all 153 splits (Table S4). The cohort-subset comparison restricted to the cohorts without a documented ALS population, shown alongside these checks in Table S4 for comparison rather than as one of the four, gave a consistent result as well. Every comparator predictor named in the Methods was run through the same procedure and is reported in full in Table S6; the regularised linear discriminant computed from the identical features agreed with the primary score to within 0.007 on most summary columns, the two exceptions being the per-cohort slope range and the between-cohort MAE SD, so the transportability result reflects how separable the calibration data are rather than the classifier family used to measure it.

Discussion

Across 18 source-study cohorts, the discriminability of a classifier fitted to a session's calibration block was related to that same session's online spelling accuracy, with a positive point estimate in every cohort, at the level of individual participants as well as between cohorts, and when the calibration recording came from an earlier session. A fitted mapping from that score to an expected accuracy did not transport. The calibration intercept had a between-cohort standard deviation of tau = 0.87 (95% CI 0.60 to 1.51) and an interval for an unrepresented cohort running from -1.97 to 1.85 on the log-odds scale; slope heterogeneity led to the same conclusion, at tau = 0.43 (0.30 to 0.77) and a range of 0.185 to 2.185. In one cohort the score estimated accuracy less well than the development-mean benchmark, and in six less well than that cohort's own mean. The association appeared in every cohort examined, though with widely varying magnitude; the numerical mapping between the score and expected accuracy did not.

That distinction determines what a calibration score can be used for. Ranking sessions within a setting, which requires only that the association hold locally, is supported. Reporting an expected accuracy in a cohort where the mapping was not developed is not supported without local recalibration, because the interval for a cohort outside this archive spans 0.032 to 0.254 for estimation error and includes chance-level discrimination and negative skill.

Mainsah and colleagues derived speller accuracy analytically from a calibration-derived detectability index and validated it within study.[15] The present analysis is the transportability counterpart to that work rather than a replacement for it: the relationship they described is reproduced here with a positive point estimate in every cohort, and what is added is the finding that the numerical mapping between the two quantities is cohort-specific. Earlier reports relating calibration measures to P300 performance in ALS[11,12] and to brain-computer interface performance more generally[10] are likewise consistent with the association reported here.

Predictor precision differs between cohorts: the number of calibration files, surviving epochs and cross-validation folds differ, and the score's standard error ranges from 0.008 to 0.031, a fourfold spread. Measurement error in a predictor attenuates a fitted slope, so some between-cohort spread in calibration could reflect precision rather than a true difference in the relationship. Correcting each cohort's slope for its own reliability, which ran from 0.848 to 0.992, moved the between-cohort standard deviation only from tau = 0.432 to 0.422, surviving a stress test that inflated the assumed measurement-error variance up to the point of losing identifiability (supplement, S9). Measurement error in the predictor is therefore real but minor, and does not explain the mapping's failure to transport.

Part of that variation is attributable to a protocol difference rather than to the cohorts themselves: the time each cohort took over a selection, the archive's only trace of the stopping rule, was associated with the two-thirds reduction in residual between-cohort variance reported above (Results, Protocol Descriptors as Moderators). That association does not restore transportability (a mapping whose slope depends on the stopping rule still cannot be carried to a site whose stopping rule is unknown), but it does change how the residual spread should be read: it is not an irreducible property of the population, and a site that reports its stopping rule alongside a calibration score would remove a substantial share of the uncertainty in a transported estimate.

The 18 development folds that produced these estimates are themselves stable. Their intercept and slope coefficients, tabulated in full in the supplement (Table S9), vary by a coefficient of variation of 2.5% and 3.3% respectively across folds, against 49.4% for the held-out cohort-specific validated slopes in Table S11. All three figures use the population standard deviation over the mean rather than the sample standard deviation reported elsewhere in this section, so they should not be reconciled arithmetically against the 0.560 figure above. The between-cohort spread this paper reports therefore reflects how each fold's mapping performs in the cohort withheld from it, not instability in what each fold fits. That distinction is evidence against development-fit noise as an alternative explanation for the heterogeneity.

The 18 folds are also not independent of one another, since any two share up to 16 of 17 development cohorts, and a joint bootstrap addresses that correlation directly rather than bounding it. Resampling every cohort's participants once per replicate and refitting all 18 leave-one-study-out folds from that single resampled draw, so the same resampled development data enter every fold within a replicate instead of being redrawn independently fold by fold, gave a within-replicate between-cohort standard deviation of the slope that averaged 0.79 across 2,000 replicates (95% range 0.49 to 1.56) and of the intercept 1.62 (0.94 to 3.36), larger in both cases than the meta-analytic tau reported above (0.43 slope, 0.87 intercept). That direction is expected rather than discordant: because this procedure never resamples which 18 cohorts are observed, only the participants within each fixed cohort, the spread it measures mixes genuine between-cohort heterogeneity with each fold's own within-cohort sampling noise and estimates something closer to the square root of tau squared plus the mean within-cohort sampling variance, which is structurally at least as large as tau. It is reported here as an empirical, assumption-light quantity that combines both sources of spread, not as a corrected or dependence-aware version of tau itself (supplement, S4).

The four ALS cohorts alone gave a more favourable and more homogeneous picture than the full archive (Results, Amyotrophic Lateral Sclerosis Subgroup); evaluating a mapping on a small number of withheld cohorts should be expected to understate how much performance varies elsewhere.

One analysis improved transportability. Excluding records in which more than 20% of calibration epochs exceeded the artifact threshold reduced estimation error from 0.090 to 0.081 and reduced the uncorrected between-cohort standard deviation of the calibration slope from 0.560 to 0.465. Screening calibration data quality before relying on a calibration-derived estimate is therefore a candidate step that warrants prospective evaluation, and it is available at no cost because the rejection fraction is computed while the score is computed.

A mapping developed entirely without ALS data estimated accuracy in the ALS cohorts with a mean absolute error of 0.108, against 0.104 when other ALS cohorts were available. The relationship is not specific to the clinical population in the sense of being absent elsewhere.

Whether it is population-dependent in a different sense is not resolved here. The calibration slope averaged higher in the four ALS cohorts than in the other 14, which would be one mechanism by which a single fitted mapping mis-calibrates, but with the cohort as the unit of analysis that difference is imprecise, the three tests of it fall on either side of the conventional threshold, a single cohort moves it, and cohort type cannot be separated from the paradigm, hardware and stopping rule that differ alongside it.

Study Limitations

First, calibration and online blocks come from the same recording session throughout. Every timestamp in the archive is de-identified, so temporal precedence within a session cannot be verified from the data, and the separation enforced here is between protocol phases rather than demonstrated ordering. The analysis using a preceding session's calibration recording is the closest available approximation and shows an attenuated association.

Second, the predictor is not the decoder that produced the outcome. It is the out-of-fold discriminability of a classifier fitted here on each session's calibration recordings, whereas the online selections were produced by a decoder fitted during the original experiment, described in the source-study publications as a linear classifier trained on that session's calibration data and, in several cohorts, coupled to a dynamic stopping or adaptive stimulus-selection rule.[32,33] The archive names no online classifier for any source study and withholds the BCI2000 parameter files, so the deployed decoder cannot be reconstructed and the two cannot be compared directly. Both are fitted on the same recordings, so the quantity evaluated is the learnability of that session's calibration data, measured against the online accuracy recorded in the same session, rather than an independent physiological marker of user aptitude.

Third, the cohorts differ in speller matrix, stimulus paradigm, electrode type, and stopping rule, and paradigm is largely nested within source study, so withholding a cohort also withholds its paradigms. Cohort and paradigm effects cannot be separated in this design. Stimulus presentation and stopping rule are known to change speller accuracy substantially,[32,34,35] so some of the between-cohort variability reported here is attributable to protocol rather than to population. The archive records none of these rules as fields, but two of them leave a recoverable trace that was tested here: the time taken per selection (the signature of the stopping rule; Results, Protocol Descriptors as Moderators), and the range of target indices, which bounds the speller alphabet from below and was not associated with a share distinguishable from zero. The archive does document a grid size and a stimulus paradigm per study, unlike the stopping rule; both were tested directly and neither was associated with a share distinguishable from zero either (Results, Protocol Descriptors as Moderators). What remains untestable is the stopping algorithm and the thresholds it used, the exact grid layout beyond its documented total size, and the online decision procedure, so the protocol explanation is now partly quantified rather than either excluded or merely acknowledged.

Fourth, 30.9% of records were at 100% accuracy and three cohorts were near ceiling, where there is little variation to estimate. The sensitivity analysis restricted to cohorts with meaningful outcome variance gave a higher estimation error, so the primary estimate is favourably influenced by cohorts in which the task was easy.

Fifth, the outcome is character-level selection accuracy reconstructed from archived event traces. It is an operational endpoint and not communication effectiveness, quality of life, or any clinical outcome, and it does not capture communication rate, for which accuracy alone is known to be an incomplete summary.[36] Sixth, these are legacy protocols, and the analysis does not establish performance with contemporary assistive-communication workflows. Seventh, the archive documentation identifies an ALS population for four cohorts only; the remaining cohorts are described as other cohorts because the documentation does not support a positive characterisation, and no participant-level clinical characteristics were available for the cohort description.

Eighth, the comparison of calibration slope between cohort types carries two further limitations of its own. Each cohort's slope comes from a model developed on the other 17 cohorts, 14 of which carry no documented ALS population, so part of the higher slope in the ALS cohorts could reflect the composition of the development set rather than the population. Those 18 slopes also share most of their development data with one another, which correlates them positively, so all three tests report standard errors that are somewhat optimistic. Neither can be addressed at this number of cohorts, and both are reasons to treat the comparison as a question for a future design rather than an answer from this one.

Ninth, the 18 held-out estimates are not fully independent: each development fold shares 16 of the other 17 cohorts with every other fold, so the between-cohort variance estimator sees correlated inputs rather than 18 independent draws. The fold-coefficient stability reported above bounds how much this can be inflating the observed spread, and the joint bootstrap in the Discussion measures it directly rather than only bounding it. Coefficients that varied widely across folds despite that shared training data would themselves be evidence of fold-to-fold instability, and they do not. That correlation is now measured rather than merely bounded: the joint bootstrap's correlation matrix (supplement, S4) shows it concentrated in a handful of cohort pairs rather than spread evenly across all 153, consistent with a small number of cohorts having outsized influence on each other's development fit rather than every fold moving in lockstep.

Tenth, the between-cohort variance is itself estimated with limited precision: eighteen cohorts of 5 to 24 participants leave tau imprecise, and a cluster-robust variance at these cluster counts is downward-biased, inflating tau and I^2 in the direction that favours the conclusion. The conclusion holds at tau's lower confidence limit, with within-cohort variances inflated fivefold (Results, Transportability), and under a participant-cluster bootstrap sensitivity analysis that does not rely on the sandwich standard-error approximation, which gave a smaller tau of 0.37 for the slope and 0.77 for the intercept (supplement, S4).

Eleventh, because participant identifiers are study-scoped, we could not determine whether any individuals participated in more than one source study.

Conclusion

In 18 source-study cohorts, calibration-derived decoder discriminability was related to online spelling accuracy in the corresponding session, with a positive point estimate in every cohort and at the level of individual participants, but the fitted mapping between the two did not transport: the calibration intercept spanned mappings that badly understate and badly overstate accuracy, the slope varied more than tenfold, and the prediction interval for a cohort outside the archive does not exclude discrimination at chance or skill at or below zero. The score may be useful for within-setting ranking after local validation, and the artifact-rejection fraction is a candidate data-quality screen that warrants prospective evaluation; neither use has been prospectively validated here, and the score should not be used to report an expected accuracy in a cohort where the mapping was not developed without local recalibration. Prospective evaluation would need to predefine the recalibration procedure and measure user-centred communication outcomes rather than character accuracy alone.

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Author contributions: Contributions are described using the CRediT taxonomy. A.G.: conceptualization, methodology, software, formal analysis, data curation, validation, visualization, and writing of the original draft. Y.A.: software, data curation, formal analysis, validation, and review and editing of the manuscript. M.O.: methodology, software, validation, and review and editing of the manuscript. Y.B.: methodology, validation, and review and editing of the manuscript. E.K.: conceptualization, methodology, supervision, resources, and review and editing of the manuscript. O.D.: conceptualization, methodology, investigation, formal analysis, clinical interpretation, supervision, writing of the original draft, and review and editing of the manuscript. All authors critically reviewed the manuscript and approved the final version submitted for publication. A.G. (corresponding author) had full access to all data in the study and takes responsibility for the integrity of the data and the accuracy of the analysis.

Data and Code Availability

BigP3BCI version 1.0.0 is publicly available (doi:10.13026/0byy-ry86). Analysis code and frozen outputs are available at https://github.com/Alon-Gorenshtein/study_bigp3_als_calibration (commit bb07fb994b0d776101afaef9f0f3dd9c12596124). The archive is not redistributed.

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